Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	20 February 2026
Team ID	LTVIP2026TMIDS88068
Project Name	Online Payments Fraud Detection using Machine learning
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

- The system should collect and process transaction data (amount, time, location, device, etc.).
- It should preprocess and clean the dataset before model training.
- The system must train a machine learning model using historical transaction data.
- It should classify transactions as **Fraud** or **Legitimate**.
- The system should generate a fraud probability or risk score.
- It must detect suspicious transactions in real time.
- The system should trigger alerts for high-risk transactions.
- It should store transaction results for future analysis and retraining.
- The model should support periodic retraining with updated data.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

- The system must provide high accuracy in fraud detection.
- It should minimize false positives and false negatives.
- The response time must be fast (real-time or near real-time processing).
- The system should be scalable to handle large volumes of transactions.
- It must ensure data privacy and security.
- The solution should be reliable and available with minimal downtime.
- The model should be maintainable and easy to update.
- The system should be user-friendly and easy to integrate with payment platforms.